R-107, First Floor, Metro Tower, Vijay Nagar, Indore (M.P.) 452010

Cypher Analytica — Power BI for Business Intelligence and Data Analytics

Level: Beginner to Advanced
Duration: 60 instructor-led hours + 30 hours of independent practice
Suitable for: Students, aspiring data analysts, business analysts, MIS professionals, managers, and business owners
Prerequisites: Basic Excel knowledge and familiarity with tabular data. SQL knowledge is helpful but not mandatory.

Teaching approach: Concept explanation, guided demonstrations, business exercises, troubleshooting, and a complete reporting project.

Course objectives

By the end of the course, learners should be able to:

- Connect to data sources and prepare data using Power Query.
- Design reliable semantic models with appropriate relationships.
- Create DAX measures for business calculations.
- Build interactive, accessible reports.
- Publish content and understand refresh, sharing, and security.
- Validate report results and troubleshoot common issues.

Module-wise curriculum

Module	Topics covered	Hours
1. Business Intelligence and Power BI Fundamentals	BI concepts; reporting workflow; Power BI Desktop and Service; reports versus dashboards; semantic models; interface navigation; understanding business requirements and KPIs	3
2. Data Connectivity and Profiling	Connecting to Excel, CSV, folders, and SQL databases; Import versus DirectQuery concepts; storage-mode tradeoffs; column quality, distribution, and profiles; source structure; identifying data-quality issues	4
3. Data Preparation with Power Query	Data types; missing values and errors; duplicate handling; text cleaning; mixed date formats and locale; split and merge columns; conditional and custom columns; append	8

Module	Topics covered	Hours
	and merge queries; pivot and unpivot; Group By; combining files; parameters; M language introduction	
4. Data Modeling and Relationships	Fact and dimension tables; defining table grain; star schema; keys; cardinality; filter direction; active and inactive relationships; date tables; role-playing dimensions; many-to-many risks; avoiding ambiguous filter paths	6
5. DAX Fundamentals	Measures versus calculated columns and tables; syntax; variables; aggregation functions; DIVIDE, IF, SWITCH, and COALESCE; COUNTROWS, DISTINCTCOUNT; iterator functions including SUMX and AVERAGEX; introductory row and filter context	6
6. DAX Context and Advanced Calculations	CALCULATE; context transition; FILTER; ALL, REMOVEFILTERS, and KEEPFILTERS; VALUES and SELECTEDVALUE; RELATED and RELATEDTABLE; USERELATIONSHIP; introductory TREATAS; ranking; percentage of total; weighted averages; virtual-table concepts	7
7. Date Analysis and Time Intelligence	Calendar requirements; year, quarter, month, and week analysis; MTD, QTD, and YTD; previous-period comparisons; DATEADD, SAMEPERIODLASTYEAR, and TOTALYTD; year-over-year growth; rolling periods; fiscal calendar considerations	5
8. Visualizations and Interactive Report Design	Selecting appropriate visuals; cards, tables, matrices, and charts; slicers; visual interactions; drill-down and drillthrough; report-page tooltips; conditional formatting; bookmarks; navigation buttons; field parameters; accessibility and mobile layouts	7
9. Power BI Service, Refresh, and Security	Publishing; workspaces; reports and dashboards; apps; sharing concepts; permissions; row-level security and role testing; gateway purpose; credentials; scheduled refresh;	5

Module	Topics covered	Hours
	refresh-history troubleshooting; licensing and organizational access considerations	
10. Validation, Performance, and Capstone Assessment	Reconciling totals; checking relationship behavior; debugging DAX; Performance Analyzer; unnecessary columns and high cardinality; query-folding concepts; capstone completion; documentation; presentation and viva	9
Total	**Guided instruction, practical work, and assessment**	**60**

Practical exercises

Learning area	Suggested task
Data profiling	Identify missing values, inconsistent types, and duplicate records
Power Query	Combine monthly sales files and standardize mixed date formats
Data modeling	Build a star schema using sales, customer, product, and date tables
DAX fundamentals	Calculate revenue, profit, margin, order count, and average order value
Filter context	Compare selected-category sales with overall sales
Relationships	Analyze orders by order date and delivery date
Time intelligence	Calculate YTD sales, prior-year sales, and growth percentage
Report interactivity	Build drillthrough pages and contextual tooltips
Security	Implement and test region-based access
Refresh troubleshooting	Diagnose source-path, credential, and gateway issues
Validation	Reconcile report totals against source records and explain differences

Capstone project options

1. **Sales Performance Report:** Revenue, profit, margin, target achievement, monthly trends, regional performance, and product contribution.
2. **HR Analytics Report:** Headcount, hires, exits, department distribution, and attrition using clearly defined reporting periods.
3. **Inventory and Supply Chain Report:** Stock movements, stock availability, supplier performance, reorder indicators, and delivery delays.
4. **Institute Performance Report:** Admissions, course enrollment, attendance, fee collections, outstanding payments, and batch performance.
5. **Financial Performance Report:** Income, expenses, budget variance, profitability, and period comparisons.

The project should develop progressively throughout the course, with final refinement and assessment in Module 10.

Mandatory project deliverables

- A Power BI Desktop project file with documented data sources.
- A data dictionary and model diagram.
- Repeatable Power Query transformations.
- At least 12 meaningful DAX measures.
- A three-page report: executive overview, detailed analysis, and drillthrough.
- Appropriate filters, navigation, tooltips, and accessible formatting.
- Validation checks for key measures and totals.
- A refresh plan and a tested security role.
- A published report where suitable account access is available; otherwise, a documented publishing walkthrough.

Assessment structure

Component	Weight
Power Query and data preparation assignments	20%

Component	Weight
Data modeling and DAX practical assessment	25%
Report design and analytical interpretation	20%
Capstone project	25%
Technical viva and troubleshooting	10%
Total	**100%**

Certification Criteria: Minimum 60% overall, completion of mandatory assignments, and successful demonstration of a validated capstone report.

Interview preparation

Practice explaining row context versus filter context, measures versus calculated columns, star schemas, relationship direction, CALCULATE, iterator functions, date-table requirements, incorrect totals, refresh failures, and row-level security.